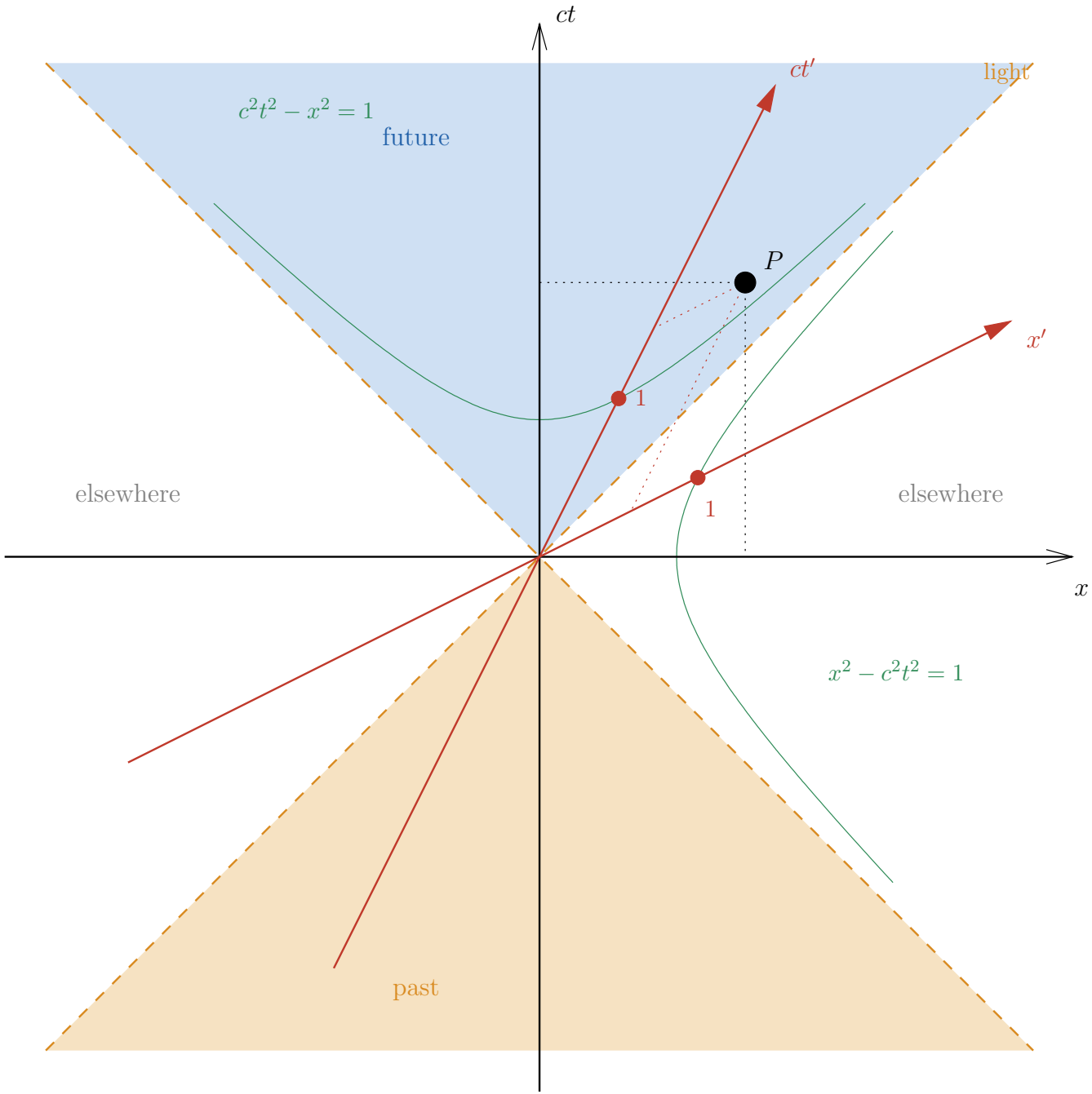
Minkowski diagram

Light cone and the axes of a frame moving at $v = 0.5\,c$; the same event has coordinates in both frames



Frame S' moves at $v = 0.5\,c$

$\beta = v/c = 0.5,\ \gamma = 1.155$

ct' axis: $x = \beta\, ct$ (tilt 26.57°)

x' axis: $ct = \beta x$

Event $P = (x, ct) = (1.5, 2.0)$

in S': $x' = 0.58$, $ct' = 1.44$

$(ct')^2 - x'^2 = (ct)^2 - x^2$

invariant interval: 1.75